

Stat 435 Lecture Notes 7

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Nonlinear regression models

Overview

- Linear models are simple (e.g., easy to fit and interpret) but quite inflexible, since often linearity assumption is a poor approximation
- We need to look beyond linear models, and choose models that somehow are not too complicated but flexible enough to considerably improve upon linear models
- We will discuss four types of models: *polynomial regression*, *spline models*, and *generalized additive models*

Polynomial regression

Model

Scalar predictor X and scalar response Y :

- Population version:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_d X^d + \varepsilon, \mathbb{E}(\varepsilon) = 0$$

- Data version:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i, \mathbb{E}(\varepsilon_i) = 0$$

- $\beta_0, \dots, \beta_d$ can be estimated by the least squares method, by pretending that x_i^j for each $j \in \{1, \dots, d\}$ is a predictor when the associated design matrix does not behave badly

Boundary effect

- Usually $d = 3$ or 4 is set since the polynomial

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d$$

may behave wildly

- Reason: dominant term $\beta_d X^d$ (when $\beta_d \neq 0$) varies wildly near boundary of observed X variable
- **Caution:** Boundary effect is a common problem for complicated models, and there are strategies to mitigate this

Estimated model

- Estimate of f :

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 + \cdots + \hat{\beta}_d x^d$$

- Covariance matrix of $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_d)^T$ as $\mathbf{C}$
- $\mathbf{C}$ can be complicated

Confidence band

- Variance of $\hat{f}(x)$ for a fixed x is

$$\begin{aligned} \text{Var} [\hat{f}(x)] &= \underbrace{(1, x, x^2, \dots, x^d)}_{l_d^T(x)} \mathbf{C} \underbrace{(1, x, x^2, \dots, x^d)^T}_{l_d(x)} \\ &= l_d^T(x) \mathbf{C} l_d(x) \end{aligned}$$

- Envelope for $\hat{f}(x)$ formed by:

$$\hat{f}(x) \pm 2\sqrt{\text{Var} [\hat{f}(x)]}$$

Illustration: model

- Model family:

$$\Pr(y_i > 250 | x_i) = \frac{\exp(\theta_i)}{1 + \exp(\theta_i)}$$

with

$$\theta_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d$$

- $d = 1$ gives simple logistic regression

Illustration: fitted

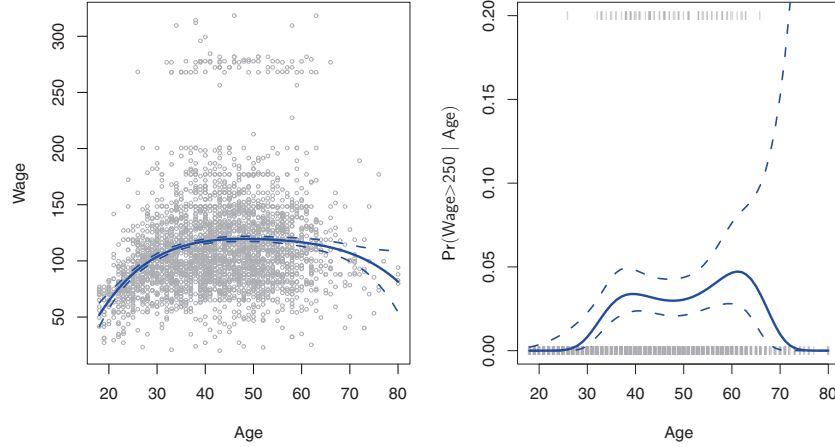


FIGURE 7.1. The *Wage* data. Left: The solid blue curve is a degree-4 polynomial of *wage* (in thousands of dollars) as a function of *age*, fit by least squares. The dotted curves indicate an estimated 95 % confidence interval. Right: We model the binary event *wage*>250 using logistic regression, again with a degree-4 polynomial. The fitted posterior probability of *wage* exceeding \$250,000 is shown in blue, along with an estimated 95 % confidence interval.

Regression splines

Basis functions

- $(d + 1)$ monomials $1, X, X^2, \dots, X^d$ generate polynomials $\mathcal{P}_d$ of degree d with elements:

$$f_d(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_d x^d$$

- d functionally linearly independent functions $f_1, \dots, f_d$ generate family $\mathcal{F}$ with elements

$$\tilde{f}(x) = \beta_0 + \beta_1 f_1(x) + \beta_2 f_2(x) + \dots + \beta_d f_d(x) \quad (1)$$

- $\{f_i\}_{i=1}^d$ are called “basis functions” for $\mathcal{F}$
- Note: these are concepts in linear algebra

Piecewise polynomials

- A *global* polynomial regression model has intrinsic disadvantages
- **Alternative strategy:** fit different polynomial regressions on different parts of data, by requiring these polynomials to be consistent in some sense, so that the *resultant model displays some stability as it transits from one part of data to the other*
- Consider the model M :

$$Y = \begin{cases} \beta_{01} + \beta_{11}X + \beta_{21}X^2 + \beta_{31}X^3, & X < c \\ \beta_{02} + \beta_{12}X + \beta_{22}X^2 + \beta_{32}X^3, & X \geq c \end{cases}$$

Piecewise polynomials

- Data version

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3, & x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3, & x_i \geq c \end{cases}$$

- M has two sub-models, M_1 and M_2 , on two distinct parts, $D_1 = \{(X, Y) : X < c\}$ and $D_2 = \{(X, Y) : X \geq c\}$, of data space
- c for X is where M transits from M_1 to M_2 (or vice versa); c is called a “knot”

Splines: I

Splines

- Splines are a type of regression polynomials:
 - They employ polynomials up to a fixed degree
 - Where two polynomials meet, some constraints are satisfied
- *Cubic splines* employ deg(3) polynomials
- *Natural splines* also satisfy linear constraints at boundary
- **Smoothing splines** directly satisfy smoothness constraints and need not to have polynomial components

Cubic splines

- Two cases of transition of M from M_1 to M_2 :
 - Case 1: M_1 does not match M_2 at c ; e.g., M jumps from M_1 to M_2 at c
 - Case 2: M_1 matches M_2 at c in some sense; e.g., M_1 matches M_2 at c , in value, (and/or) in speed of change, (and/or) in curvature
- For practical purposes it is often desirable to have Case 2, and the 3-degree polynomials with matching values (i.e., M itself is continuous), speeds (i.e., first-order derivatives) and curvatures (i.e., second-order derivatives) at each knot are called **cubic splines**

Illustration

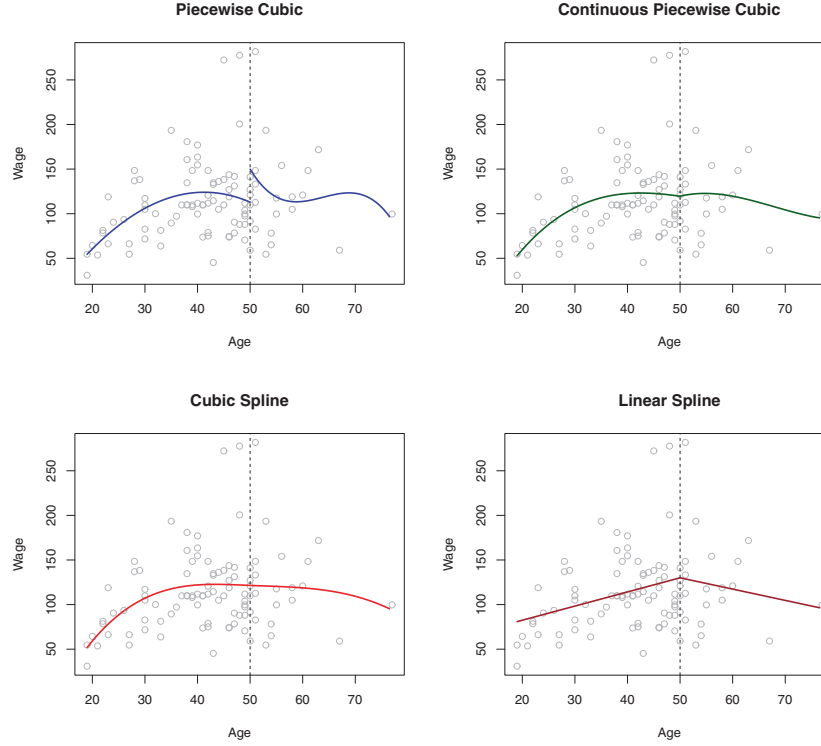


FIGURE 7.3. Various piecewise polynomials are fit to a subset of the *Wage* data, with a knot at *age*=50. Top Left: The cubic polynomials are unconstrained. Top Right: The cubic polynomials are constrained to be continuous at *age*=50. Bottom Left: The cubic polynomials are constrained to be continuous, and to have continuous first and second derivatives. Bottom Right: A linear spline is shown, which is constrained to be continuous.

Smoothing splines

- Split the range of X into $K + 1$ parts and only employ polynomials of degree d ; so needing K knots and $K + 1$ such polynomials
- Restrict these polynomials such that there are s constraints for each distinct pair of them at a knot, then we have $p = T - C$ parameters to estimate
- $T = (K + 1) \times (d + 1)$; $(K + 1)$ number of distinct regions of values of X , and $d + 1$ number of parameters in each polynomial
- $C = K \times s$; K number of knots and s number of constraints at each knot

Smoothing splines

In the above formula, s , the number of constraints at each knot can be understood as follows:

- If submodels M_i and M_{i+1} , $i = 1, \dots, K$ match in their values and up to their $(s - 1)$ -th order derivatives at each knot Q_i , then at knot Q_i there are s constraints on their associated polynomials

- If we take 0-th order derivative as performing no differentiation but keeping original function untouched, then M_i and M_{i+1} , $i = 1, \dots, K$ matching up to their r -th order derivatives at each knot Q_i induces $r + 1$ constraints

Smoothing splines: representation

- Consider $d = 3$. A cubic spline with K knots can be modelled as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \dots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i$$

for an appropriate choice of basis functions $b_1, b_2, \dots, b_{K+3}$

- One such basis is partially induced by the function

$$h(x, \xi) = (x - \xi)_+^3 = \begin{cases} (x - \xi)^3, & x > \xi \\ 0, & x \leq \xi \end{cases}$$

Smoothing splines: representation

- In summary, to fit a cubic spline to a data set with K knots, we can equivalently fit the model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 \underbrace{h(X, \xi_1)}_{\text{the 4th predictor}} + \dots + \beta_{K+3} h(X, \xi_K) + \varepsilon$$

or its data version

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \beta_4 h(x_i, \xi_1) + \dots + \beta_{K+3} h(x_i, \xi_K) + \varepsilon_i$$

where $\xi_1, \dots, \xi_K$ are the knots

- Fitting this model uses $K + 4$ degrees of freedom. We can use the least squares method to fit the model

Natural splines

- As for any regression that employs a polynomial, fitted model may be unstable near or at the boundary (induced by observed values of predictors)
- A **natural spline** is a regression spline which is required to be *linear at the boundary (in the region where X is smaller than the smallest knot, or larger than the largest knot)*

Natural/Cubic spline

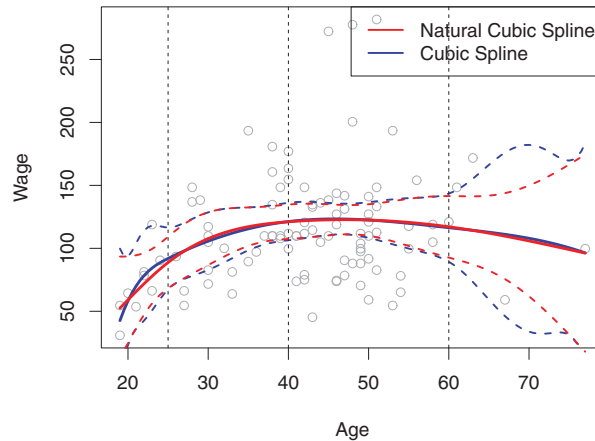


FIGURE 7.4. A cubic spline and a natural cubic spline, with three knots, fit to a subset of the *Wage* data.

- Each cubic spline has 3 knots

Splines: II

More knots?!

- Regression spline is most flexible in regions that contain a lot of knots, because in those regions the polynomial coefficients can change rapidly. Hence, one option is to place more knots in places where we feel the function might vary most rapidly, and to place fewer knots where it seems more stable
- While this option can work well, in practice it is common to place knots in a uniform fashion. One way to do this is to specify the desired degrees of freedom, and then have the software automatically place the corresponding number of knots at uniform quantiles of the data”, as done, e.g., by 10-fold CV based on the residual sum of squares (RSS) values

Placing knots

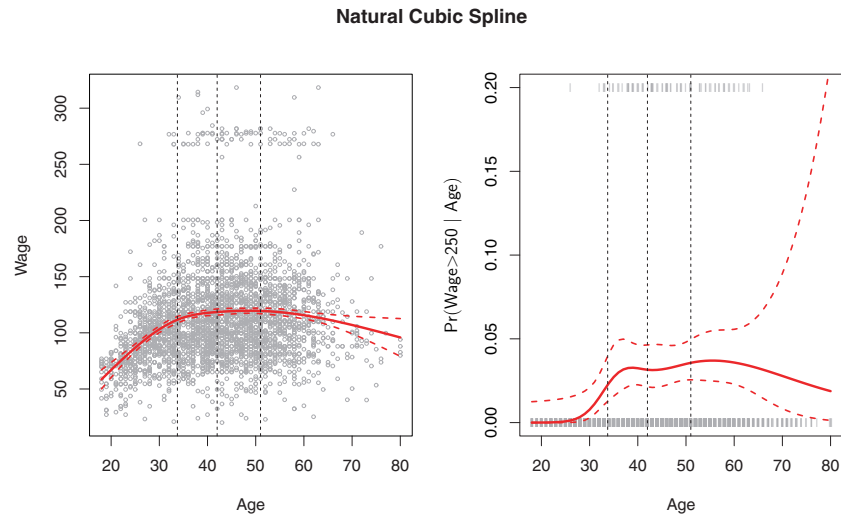


FIGURE 7.5. A natural cubic spline function with four degrees of freedom is fit to the `Wage` data. Left: A spline is fit to `wage` (in thousands of dollars) as a function of `age`. Right: Logistic regression is used to model the binary event `wage > 250` as a function of `age`. The fitted posterior probability of `wage` exceeding \$250,000 is shown.

- 3 knots, placed at 25th, 50th, 75th percentiles of `age`

Choosing knots

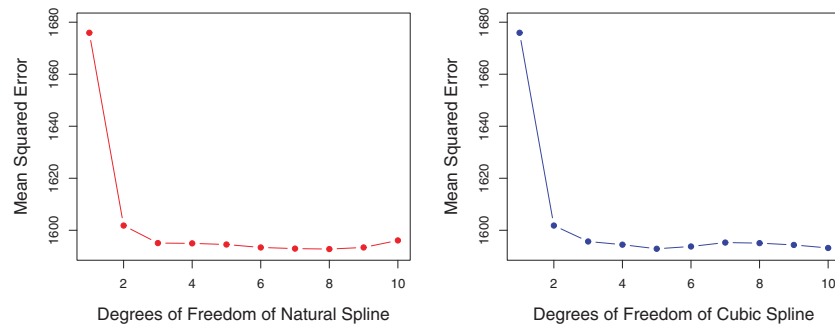


FIGURE 7.6. Ten-fold cross-validated mean squared errors for selecting the degrees of freedom when fitting splines to the `Wage` data. The response is `wage` and the predictor `age`. Left: A natural cubic spline. Right: A cubic spline.

Comparison

“Regression splines often give superior results to polynomial regression”:

- To be more flexible, polynomial regressions resort to higher degree polynomials which results in added instability at boundary (of observations for X).

- Splines only need to plan for better knots in order to achieve flexibility, and natural splines alleviate instability at boundary
- Spline captures much better (than polynomial regression) local behavior of f in true model $E(Y|X) = f(X)$ where f changes rapidly.

Illustration

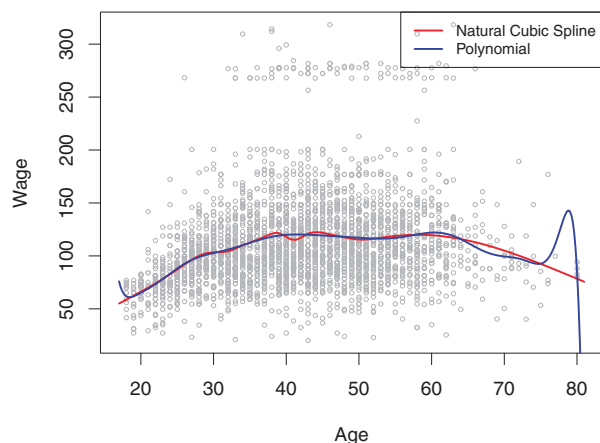


FIGURE 7.7. On the **Wage** data set, a natural cubic spline with 15 degrees of freedom is compared to a degree-15 polynomial. Polynomials can show wild behavior, especially near the tails.

Smoothing splines

Motivation

Smoothing splines are splines produced directly by restricting how **smooth** a candidate function can be:

- Given the data $\{(x_i, y_i) : i = 1, \dots, n\}$, there is a function g such that $\sum_{i=1}^n (y_i - g(x_i))^2 = 0$
- In terms of the local properties of g such as continuity and differentiability, we can restrict the “integrated curvature of g ”, which is $\int [g''(t)]^2 dt$. The integral essentially restricts how variable g'' is (and hence approximately how variable the curvature of g is)

Formulation

- A function g that minimizes

$$\underbrace{\sum_{i=1}^n (y_i - g(x_i))^2}_{\text{squared loss}} + \underbrace{\lambda \int [g''(t)]^2 dt}_{\text{penalty on smoothness}} \quad (2)$$

(for each fixed $\lambda \geq 0$) is called a *smoothing spline*”*

- When is $g''(t)$ a constant?

- For practical usage, $\deg(g) \geq 3$ if g is a polynomial
- Penalty term referred to as *roughness penalty*

Key observations

- When $\lambda = 0$, the penalty term has no effect and g can be discontinuous and will exactly interpolate the data
- When $\lambda \neq 0$, g has to be twice differentiable in order to possess a second order derivative
- When $\lambda = \infty$, the minimizer $\hat{g}$ must satisfy $\hat{g}''(t) = 0$. This means that $\hat{g}$ has to be a linear function, i.e., $\hat{g}(x) = \beta_0 + \beta_1 x$, in which case we get linear regression

Complexity and performance

The minimizer $\hat{g}_\lambda$ for $\lambda \in (0, 1)$, being a shrunken version of a natural cubic spline, has the following properties:

- $\hat{g}_\lambda$ is a piecewise cubic polynomial with knots at the unique values of $x_1, \dots, x_n$
- $\hat{g}_\lambda$ has continuous first and second order derivatives at each knot
- $\hat{g}_\lambda$ is linear in the region outside of the extreme knots

Proof of the above is a bit complicated and uses machinery from functional analysis and differential equations

Degrees of freedom

- Tuning parameter λ controls the *effective degrees of freedom*, df_λ , of $\hat{g}$
- df_λ decreases from n to 2 as λ increases from 0 to ∞
- df_λ measures flexibility of $\hat{g}_\lambda$, in the sense that larger df_λ induces a more flexible $\hat{g}_\lambda$

Degrees of freedom

- Let $\mathbf{S}_\lambda \in \mathbb{R}^{n \times n}$ be formed by evaluating associated basis functions at the x_i 's. Then

$$\hat{\mathbf{g}}_\lambda = \begin{pmatrix} \hat{g}_\lambda(x_1) \\ \hat{g}_\lambda(x_2) \\ \vdots \\ \hat{g}_\lambda(x_n) \end{pmatrix} = \mathbf{S}_\lambda \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

- **Effective degrees of freedom** is defined as

$$df_\lambda = \sum_{i=1}^n \mathbf{S}_\lambda(i, i),$$

where $\mathbf{S}_\lambda(i, j)$ is the (i, j) entry of $\mathbf{S}_\lambda$

Choose tuning parameter

- Best λ achieves the smallest cross-validation RSS among a grid values of λ
- Quick computation via *Leave-one-out-cross-validation (LOOCV)* in a single pass on the whole data:

$$RSS_{cv}(\lambda) = \sum_{i=1}^n \left(y_i - \hat{g}_{\lambda}^{(-i)}(x_i) \right)^2 = \sum_{i=1}^n \left[\frac{y_i - \hat{g}_{\lambda}(x_i)}{1 - \mathbf{S}_{\lambda}(i, i)} \right]^2$$

- $\hat{g}_{\lambda}^{(-i)}$: fitted on $\{(x_j, y_j) : j \neq i\}$
- $\hat{g}_{\lambda}$: fitted on $\{(x_i, y_i) : i = 1, \dots, n\}$

Illustration

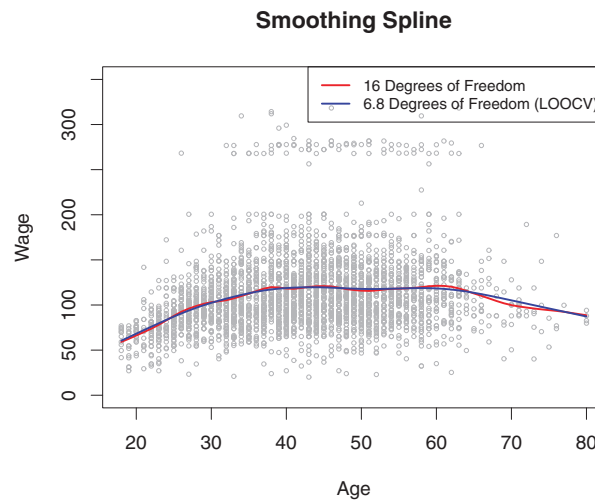


FIGURE 7.8. Smoothing spline fits to the **Wage** data. The red curve results from specifying 16 effective degrees of freedom. For the blue curve, λ was found automatically by leave-one-out cross-validation, which resulted in 6.8 effective degrees of freedom.

Generalized additive models

Overview

- *Generalized additive models (GAMs)* employs a non-linear function for each predictor while maintaining additivity of the transformed predictors
- GAMs can be applied to both quantitative and qualitative responses
- Consider

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i$$

- Replace $\beta_j x_{ij}$ by $f_j(x_{ij})$ to get a GAM as

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \dots + f_p(x_{ip}) + \varepsilon_i$$

for a set of known functions $\{f_j\}_{j=1}^p$

Illustration: I

- f_1 and f_2 natural splines; f_3 piecewise constant:

$$\text{wage} = \beta_0 + f_1(\text{year}) + f_2(\text{age}) + f_3(\text{education}) + \varepsilon$$

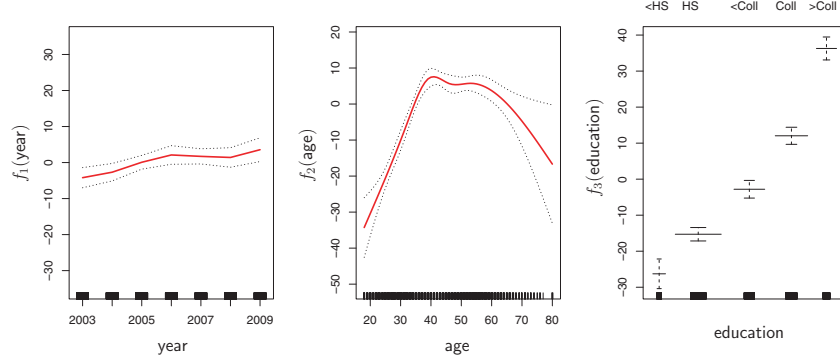


FIGURE 7.11. For the **Wage** data, plots of the relationship between each feature and the response, **wage**, in the fitted model (7.16). Each plot displays the fitted function and pointwise standard errors. The first two functions are natural splines in **year** and **age**, with four and five degrees of freedom, respectively. The third function is a step function, fit to the qualitative variable **education**.

Illustration: II

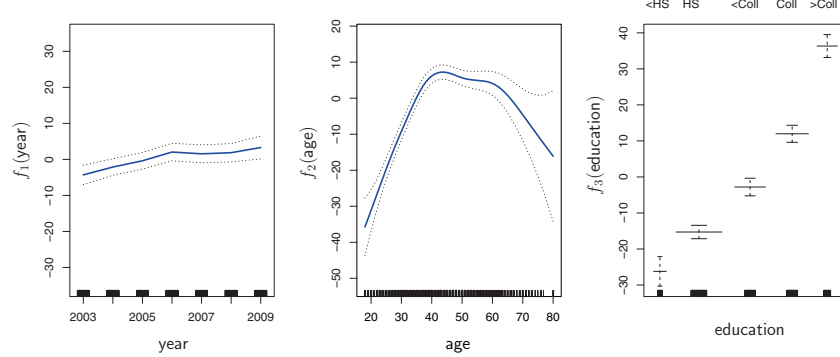


FIGURE 7.12. Details are as in Figure 7.11, but now f_1 and f_2 are smoothing splines with four and five degrees of freedom, respectively.

GAM logistic Reg

- A binary response, linear version:

$$\log \left(\frac{\Pr(Y = 1|X)}{1 - \Pr(Y = 1|X)} \right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

- GAM version:

$$\log \left(\frac{\Pr(Y = 1|X)}{1 - \Pr(Y = 1|X)} \right) = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i$$

Illustration: III

- Predict

$$p(X) = \Pr(\text{wage} > 250 | \text{year}, \text{age}, \text{education})$$

- Model:

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 \times \text{year} + f_2(\text{age}) + f_3(\text{education})$$

- Note: there are no individuals with less than a high school education who earn over \$250,000 per year

Illustration: III

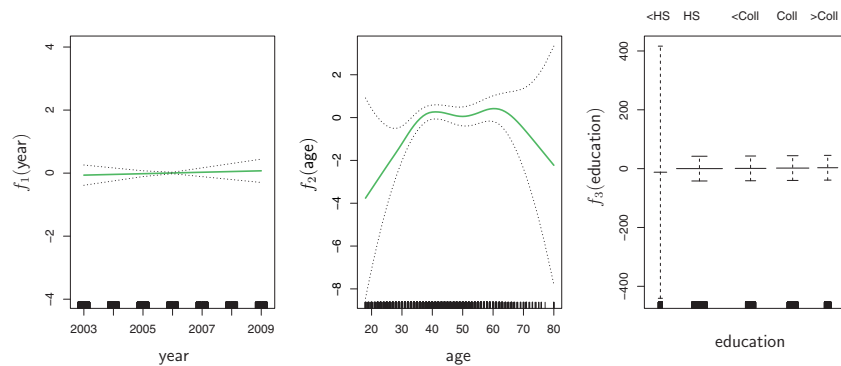


FIGURE 7.13. For the **Wage** data, the logistic regression GAM given in (7.19) is fit to the binary response $\mathbf{I}(\text{wage} > 250)$. Each plot displays the fitted function and pointwise standard errors. The first function is linear in **year**, the second function a smoothing spline with five degrees of freedom in **age**, and the third a step function for **education**. There are very wide standard errors for the first level **<HS** of **education**.

- GAMs are implemented by R library **gam**

License and session Information

License

```
> sessionInfo()
R version 3.5.0 (2018-04-23)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19044)
```

Matrix products: default

locale:

```
[1] LC_COLLATE=English_United States.1252
[2] LC_CTYPE=English_United States.1252
[3] LC_MONETARY=English_United States.1252
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.1252
```

attached base packages:

```
[1] stats      graphics  grDevices utils      datasets  methods  
[7] base
```

other attached packages:

```
[1] knitr_1.21
```

loaded via a `namespace` (and not attached):

```
[1] compiler_3.5.0  magrittr_1.5    tools_3.5.0  
[4] htmltools_0.3.6 yaml_2.2.0      Rcpp_1.0.12  
[7] stringi_1.2.4   rmarkdown_1.11 stringr_1.3.1  
[10] xfun_0.4        digest_0.6.18   evaluate_0.13
```